This assignment will not be graded and is only for practice.

Level 1

1) Rank of the Hessian matrix $H_f(x, y, z)$ of the function $f(x, y, z) = 3x^2 + xz + 2zy - z^2$ at point $(1, 0, 1)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

2) Choose the correct statements about $f(x, y, z) = 3x^2z + 2z - 4zy^2$.

1 point

- [ ] $(0, 0, 0)$ is a critical point of f.
- [ ] Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8z & -8y \\ 6x & -8y & 0 \end{bmatrix}$.
- [ ] Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8y & -8z \\ 6x & -8z & -8y \end{bmatrix}$.
- [ ] Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 3.
- [ ] Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8z & -8y \\ 6x & -8y & 0 \end{bmatrix}$.

Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 3.

Consider the following function

$$f(x, y, z) = x^3 + xy^2 + x^2 + y^2 + 3z^2$$

Answer questions 3,4,5 and 6 based on the function f.

3) Number of critical points of f is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Rank of the Hessian matrix of f at point $(\frac{-1}{3}, 0, 0)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Rank of the Hessian matrix of f at point $(0, 0, 0)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Level 2

6) Choose the correct statements.

1 point

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$(\frac{-1}{3}, 0, 0)$ is a saddle point of f.

- [ ] $(\frac{-2}{3}, 0, 0)$ is a saddle point of f.
- [ ] $(\frac{-2}{3}, 0, 0)$ is a local minimum of f.
- [ ] $(0, 0, 0)$ is a local minimum of f.
- [ ] $(0, 0, 0)$ is a local maximum of f.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{-2}{3}, 0, 0)$ is a saddle point of f.
$(0, 0, 0)$ is a local minimum of f.

7) Let a denote the shortest distance from the origin to the plane $z = x + y + 3$. The value of a^2 is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

8) Consider a function $f(x, y, z)$. Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k.Then which of the following options is(are) true? ***1 point***

- [ ] If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$, then v_1 is a saddle point.
- [ ] If $H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_2 is a saddle point.
- [ ] If $H_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_3 is a point of local maximum.
- [ ]

If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.

- [ ] If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.

- [ ] If $H_3 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then v_3 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$, then v_1 is a saddle point.

If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.

Your score is: 0/8